

Bayesian Inference Engine Visualiser

The core mechanic:

1. Human and AI start at 50/50 prior belief. They sit on opposite sides of the belief bar; as the human's overconfidence grows, it actively penalises the AI's posterior (more noise to decode = harder job).
2. Every round is two-phase: Human Speaks updates priors based on which cognitive bias they're exhibiting, then AI Responds samples a stochastic outcome from the posterior distribution.

Six human bias scenarios (Anchoring, Vague Intent, Confirmation Bias, Dunning-Kruger, Framing, and the rare "Clear Intent") each have different likelihood profiles for the three AI outcome types.

The AI outcome isn't binary; it distributes across six states: exact hit, partial hit, lucky guess, asks for clarification, close miss, badly wrong. Because you were right "the right thing" isn't one thing.

Bayesian update rule kicks in after each AI response: correct answers lift the AI's posterior, wrong answers collapse it via a likelihood ratio, and clarification is rewarded modestly as epistemically honest behaviour.

Note: made human made overconfidence deliberately self-reinforcing on wrong AI answers, they will blame the model and get more certain. That's the bit that will hurt you if you run it a few dozen times on Dunning-Kruger mode.